```
'Ke'Shawn Stafford
'Project 1
'01/28/2024

' {$STAMP BS2}
' {$PBASIC 2.5}

sensor      CON 15
servo       CON 14
LCD         CON 13
RL          CON 12
YL          CON 11
GL          CON 10
speaker     CON 9
reset       CON 1
LcdCls      CON 12
LcdCr       CON 13
LcdOff      CON 21
LcdOn       CON 22

number          VAR Byte
cursorStart     VAR Byte
head            VAR Byte
tail            VAR Byte
pointer         VAR Byte
character       VAR Byte
blinkState      VAR Bit
i               VAR Byte
counter         VAR Byte

message DATA "Theater Full! Push Reset!  "

Main:
    number = 0
    blinkState = 0
    counter = 0

    SEROUT LCD, 84, [LcdOn, LcdCls]
    PAUSE 50
    GOSUB UpdateDisplay

DO
    FOR i = 0 TO 20
        PULSOUT 14, 1190
    NEXT

      IF IN15 = 1 THEN
        PAUSE 5
        number = number + 1
        GOSUB UpdateLEDs
        ENDIF

    IF number >= 46 THEN
        counter = counter + 1
        IF counter = 5 THEN
            counter = 0
            IF blinkState = 0 THEN
                HIGH RL
                blinkState = 1
            ELSE
                LOW RL
                blinkState = 0
            ENDIF
        ENDIF
    ENDIF
```

```
    IF number = 50 THEN
        GOSUB LockDoor
        GOSUB SoundPlayed
        GOTO ScrollFullMessage
    ENDIF

    GOSUB UpdateDisplay

LOOP

UpdateDisplay:
    SEROUT LCD, 84, [128, "Nightmare Theater   "]
    SEROUT LCD, 84, [148, "No of People:", DEC2 number]
    IF number < 50 THEN
        SEROUT LCD, 84, [168, "No of Vacancies:", DEC2 (50 - number)]
    ENDIF
    RETURN

ScrollFullMessage:
DO
    SEROUT LCD, 84, [12]
    PAUSE 50
    cursorStart = 148
    head = 0
    FOR tail = 0 TO 27
        SEROUT LCD, 84, [cursorStart]

        FOR pointer = 0 TO 19
            READ (head + pointer) // 27 + message, character
            SEROUT LCD, 84, [character]
        NEXT

        head = head + 1

        IF head >= 27 THEN
            head = 0
        ENDIF

        FOR counter = 0 TO 10
        PAUSE 25

        IF IN1 = 1 THEN
        GOTO Resett
        ENDIF

        NEXT
        NEXT
     LOOP

UpdateLEDs:
    IF number <= 40 THEN
        HIGH GL
        LOW YL
        LOW RL
    ELSEIF number >= 41 AND number <= 45 THEN
        HIGH YL
        LOW GL
        LOW RL
    ELSEIF number >= 46 AND number <= 50 THEN
        LOW GL
        LOW YL
    ENDIF
    RETURN

LockDoor:
    FOR i = 0 TO 25
```

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        PULSOUT 14, 255
        PAUSE 20
    NEXT
    RETURN

SoundPlayed:
        FREQOUT speaker, 200, 1047
        FREQOUT speaker, 200, 1175
        RETURN

Resett:
    FREQOUT speaker, 200, 1568
    FREQOUT speaker, 200, 1760
    GOTO main
```